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```

```
# Note: This code assumes you run it in a PySpark environment where `sc`
(SparkContext) is available.
```

```
# If you run with `spark-submit` or in a PySpark shell, `sc` will be present.
```

```
# In a plain Python script, you need to create a SparkContext first:
```

```
# from pyspark import SparkContext
```

```
# sc = SparkContext()
```

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```
# Q1: Create an RDD
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```
numbers = list(range(1, 16))    # 1. Create a list of numbers from 1 to 15
```

```
rddA = sc.parallelize(numbers)  # 2. Convert the list into an RDD named rddA
```

Q1 Results

```
print("rddA collected:", rddA.collect())
```

```
print("Number of partitions:", rddA.getNumPartitions())
```

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```
# Q2: Apply Transformations (Map)
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```
rddB = rddA.map(lambda x: x ** 2) # Square each number
```

```
rddC = rddB.map(lambda x: x + 10) # Add 10 to each squared number
```

Q2 Results

```
print("rddB:", rddB.collect())
```

```
print("rddC:", rddC.collect())
```

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# Q3: Short Answer
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# Two examples of Transformations: map(), filter()
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# Two examples of Actions: collect(), count()
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